Q1. Briefly compare the following concepts. You may use an example to explain your point(s).

(a) Snowflake schema, fact constellation.

(b) Data cleaning, data transformation, refresh.

(c) Enterprise warehouse, data mart, virtual warehouse.

Answer:

(a) The **snowflake schema** and **fact constellation** are both variants of the **star schema model**, which consists of a fact table and a set of dimension tables; the **snowflake schema** contains some normalized dimension tables, whereas the **fact constellation** contains a set of fact tables that share some common dimension tables.

(b) **Data cleaning** is the process of detecting errors in the data and rectifying them when possible. **Data transformation** is the process of converting the data from heterogeneous sources to a unified data warehouse format or semantics. **Refresh** is the function propagating the updates from the data sources to the warehouse.

(c) **An enterprise warehouse** collects all of the information about subjects spanning the entire organization, whereas **a data mart** contains a subset of corporate-wide data that is of value to a specific group of users.

An enterprise warehouse provides corporate-wide data integration, usually from one or more operational systems or external information providers, and is cross-functional in scope, whereas **the data mart** is confined to specific selected subjects (such as customer, item, and sales for a marketing data mart).

An enterprise warehouse typically contains detailed data as well as summarized data, whereas the data in **a data mart** tend to be summarized.

An enterprise data warehouse may be implemented on traditional mainframes, computer super-servers, or parallel architecture platforms, while **data marts** are usually implemented on low-cost departmental servers that are UNIX/LINUX- or Windows-based.

The implementation cycle of **an enterprise warehouse** may take months or years, whereas that of **a data mart** is more likely to be measured in weeks.

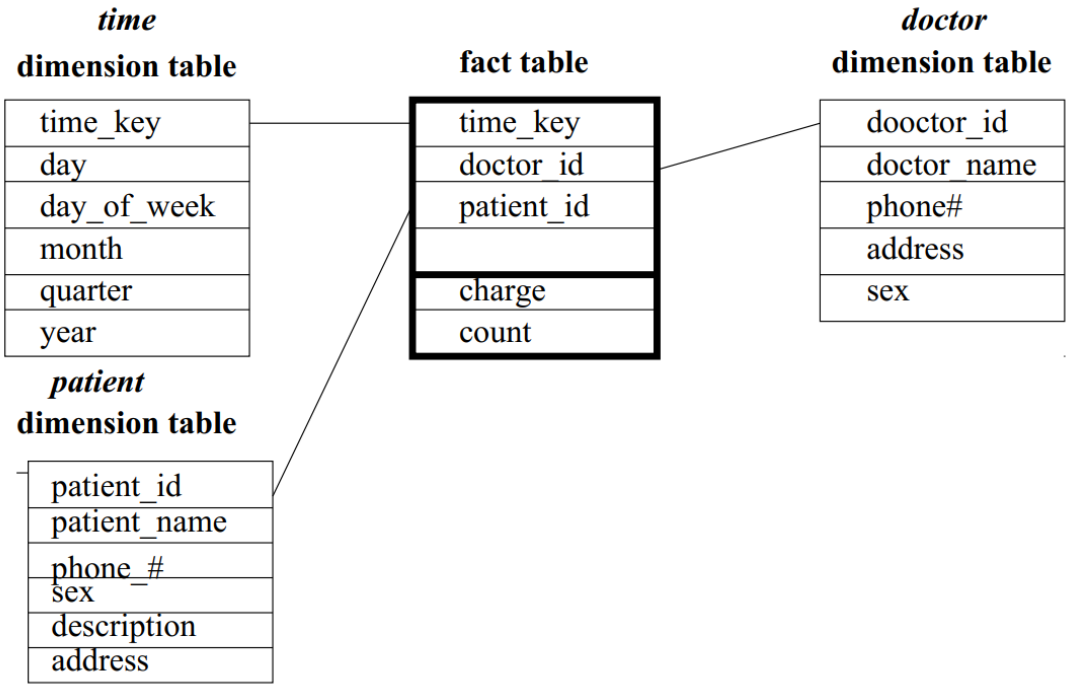
A virtual warehouse is a set of views over operational databases. For efficient query processing, only some of the possible summary views may be materialized. **A virtual warehouse** is easy to build but requires excess capacity on operational database servers.

Q2. Suppose that a data warehouse consists of the three dimensions *time, doctor*, and *patient*, and the two measures *count* and *charge*, where *charge* is the fee that a doctor charges a patient for a visit.

(a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.

Answer: The **star schema**, the **snowflake schema**, and the **fact constellations schema**

(b) Draw a schema diagram for the above data warehouse using one of the schema classes listed in (a).



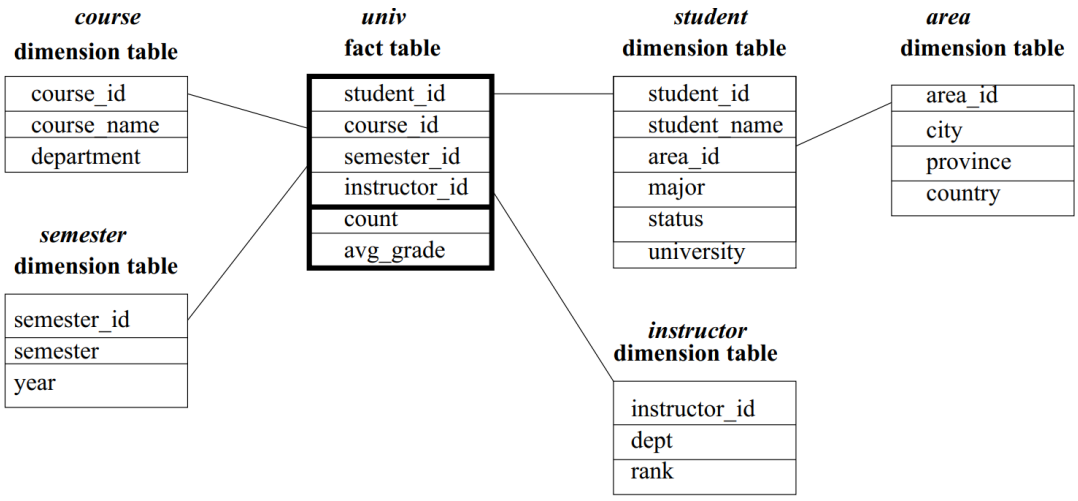
(c) Starting with the base cuboid [day; doctor; patient], what specific OLAP operations should be performed in order to list **the total fee collected by each doctor in 2025**?

Answer:

- **Roll-up** on *time* from *day* to *year*.
- **Slice** for *time=2025*.
- **Roll-up** on *patient* from *individual patient* to *all.*

Q3. Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures (count and avg grade). When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the **avg grade measure** stores the actual course grade of the student. At higher conceptual levels, the **avg grade** stores the average grade for the given combination.

- Draw a snowflake schema diagram for the data warehouse.



- Starting with the base cuboid [student; course; semester; instructor], what specific OLAP operations (e.g., roll-up from semester to year) should one perform in order to list **the average grade of CS courses for each Big University student**.
  Answer: The specific OLAP operations to be performed are:
  - **Roll-up** on course from *course_id* to *department*.
  - **Roll-up** on student from *student_id* to *university*.
  - **Dice** on course, student with *department="CS"* and *university = "Big University"*.
  - **Drill-down** on *student* from *university* to *student_name*.

- If each dimension has five levels (including all), such as "student < major < status < university < all", how many cuboids will this cube contain (including the base and apex cuboids)?
  Answer: This cube will contain 5^4 = 625 cuboids.

Q4. Answering the following questions:

- Enumerate three categories of measures, based on the kind of aggregate functions used in computing a data cube.
  Answer: The three categories of measures are **distributive, algebraic, and holistic**.

- What are the main types of data warehouse usage? And what are the differences between them?
  Answer: There are three main types of data warehouse usage: ***information processing***, ***analytical processing***, **and** ***data mining***.
  - **Information processing** involves using queries to find and report useful information using crosstabs, tables, charts, or graphs.
  - **Analytical processing** uses basic OLAP operations such as slice-and-dice, drill-down, roll-up, and pivoting on historical data in order to provide multidimensional analysis of data warehouse data.
  - **Data mining** uses knowledge discovery to find hidden patterns and associations, constructing analytical models, performing classification and prediction, and presenting the mining results using visualization tools.

- Given the data in the following table, please draw a Bitmap Index table for column “Course”.

StudentID	Course	Grade
S1	2802ICT	C
S2	3802ICT	D
S3	3804ICT	D
S4	3803ICT	E
S5	4030ICT	H

Answer: Index on Course:

RecID	2802ICT	3802ICT	3803ICT	3804ICT	4030ICT
S1	1	0	0	0	0
S2	0	1	0	0	0
S3	0	0	0	1	0
S4	0	0	1	0	0
S5	0	0	0	0	1

- What is data generalization? Why we need data generalization?
  Answer: **Data generalization** summarizes data by replacing relatively low-level values with higher-level concepts or by reducing the number of dimensions to summarize data in concept space involving fewer dimensions.

  Given the large amount of data stored in databases, it is useful to be able to describe concepts in concise and succinct terms at generalized (rather than low) levels of abstraction.